

# Sean Kim

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## EDUCATION

### University of California, Irvine

Expected June 2027

*Bachelor of Science in Computer Science*

*Irvine, CA*

**Coursework:** Object-Oriented Programming, Data Structure Implementation, Design & Analysis of Algorithms, Principles in System Design, Computer Organization, Computer Networks, Discrete Structures, Data Management, Data & Web Applications, Information Retrieval, Machine Learning & Data Mining, Linear Algebra, Probability & Statistics

## EXPERIENCE

### Software Engineer Intern

June 2025 – Sept 2025

*MIADVG*

*Irvine, CA*

- Built and shipped 3 production React and Next.js applications in TypeScript with Tailwind CSS serving 50,000+ combined users, delivering features from client requirements in 2-week Agile sprints
- Developed 10+ Express REST endpoints on Node.js over PostgreSQL and MySQL, cutting API latency 67% from 300ms to 100ms by collapsing 100+ per-item queries into a single JOIN
- Integrated third-party APIs spanning payments, SMS, and authentication, handling Stripe checkout with webhook order confirmation, Twilio messaging, and Google OAuth 2.0 sign-in
- Automated AWS EC2 deployment with a GitHub Actions CI/CD pipeline that ran builds and a Vitest unit test suite on each merge to main, blocking failed releases while shipping 2–3 releases weekly

### Software Developer Intern

Jan 2025 – May 2025

*CALIT2, University of California, Irvine*

*Costa Mesa, CA*

- Developed a Python Flask-SocketIO backend that streamed real-time detections from 5+ camera feeds to a React security dashboard over WebSockets, sustaining 100–200ms event-delivery latency
- Cut duplicate alerts by 40% on the security dashboard with a Python pipeline that deduplicated detections via a Redis TTL cache and routed them through RabbitMQ priority queues
- Fine-tuned YOLO with PyTorch transfer learning on labeled client footage, improving validation mAP by 8 points over the stock pretrained baseline powering the real-time detection pipeline

### Undergraduate Research Assistant

Jan 2024 – June 2024

*University of California, Irvine*

*Irvine, CA*

- Created Python ETL pipelines with BeautifulSoup, PyPDF2, and Pandas that cleaned and normalized thousands of UCI PDF/HTML documents into training-ready text for iGPT development
- Evaluated fine-tuned GPT-3.5 responses on UCI-specific queries to validate answer quality during early-stage iGPT development, a project that later evolved into ZotGPT for UCI students

## PROJECTS

### Docent | [GitHub] [Live] | *Next.js, Python, FastAPI, Pinecone, Voyage, Claude, Tavily*

Apr 2026 – July 2026

- Improved answers on complex questions with an agentic loop that splits each query into sub-questions, searches documents and the web, and merges them into one grounded answer
- Reduced hallucinations with a hand-built RAG pipeline (no LangChain) citing every source inline, shipped to production as Next.js and FastAPI services with per-IP rate limiting, Google sign-in, and pytest CI

### CollabCode | [GitHub] [Live] | *React, TypeScript, Monaco, Flask-SocketIO, Redis, Postgres*

Feb 2026 – Apr 2026

- Sustained 17ms p95 sync latency with 50 concurrent clients per room by building a real-time collaborative code editor with live cursors, follow mode, and multi-file workspaces
- Implemented the React sync layer for Monaco with three-layer persistence (in-memory, Redis, PostgreSQL), plus four Claude API streaming modes and sandboxed server-side Python execution

## TECHNICAL SKILLS

**Languages:** Python, JavaScript, TypeScript, Java, C++, SQL, HTML/CSS

**Tools and Frameworks:** React, Next.js, Node.js, Express, Flask, FastAPI, PostgreSQL, MySQL, Redis, Docker, AWS (EC2), Git, Linux, GitHub Actions, WebSockets, RabbitMQ, Tailwind CSS, pytest, Vitest

**AI/ML:** Retrieval-Augmented Generation (RAG), Vector search, Pinecone, Agentic pipelines, Fine-tuning, PyTorch, YOLO, Anthropic SDK